

Day 42 - SSH

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8:37 PM

SSH Info

Console Port Security

1. Login

You can configure a password on the **console line** of a Cisco IOS device by entering **console line** configuration mode and using a few commands:

- **line console 0**
 - Enters the **console line** configuration mode. There's only one single **console line**, that's why we specified a value of '0'.
- **password PASSWORD**
 - Assigns a password for the **console line** to be accessed.
- **login**
 - Enables the **login** feature, which prompts the user to enter the **console line**'s password before they can access the Cisco CLI.

2. Login Local

The **console line** can also be configured to require users to login using one of the configured usernames and passwords on the device through these commands:

- **username USERNAME secret PASSWORD**
 - Configures the username and password to be used when users log into the Cisco CLI.
- **line console 0**
 - Enters the **console line** configuration mode. There's only one single **console line**, that's why we specified a value of '0'.
- **login local**
 - Enables the **login local** feature, which prompts the user to enter a configured username and password before they can access the Cisco CLI.

Default login settings

- By default, no password is needed access the Cisco CLI of a Cisco IOS device via the **console port**.

Login & Login Local

- If both **login** and **login local** are enabled on the device at the same time, then the **login local** will take priority over the other setting and will be used for authentication.

Layer 2 Switch - Management IP

Layer 2 switches are not IP routing aware, as in they don't perform packet routing and don't build routing tables. In this situation, we cannot connect to the switch through Telnet/SSH and perform management configurations.

- However, an IP address can be assigned onto a **Switch Virtual Interface (SVI)** to allow remote connections to the CLI of the switch. (Through **Telnet/SSH**)
- To perform that, you'd have to create an **SVI** through the **interface vlan NUMBER** command, assign it an IP address and enable it through the **no shutdown** command.
- Then, you configure the switch's **default gateway** to a router in its LAN through the **ip default-gateway IP-ADDRESS** command.

The Importance of a Switch's Default Gateway

- If the switch doesn't have a **default gateway** to a router in its LAN, it cannot communicate with PCs which are outside of its LAN, therefore, **Telnet/SSH** communications cannot be made.

Telnet General Information

Teletype Network (Telnet) is an old protocol used to remotely access the CLI of a remote host.

- **Telnet** servers listen on port **23**.
- It has been replaced by the more secure protocol '**SSH**', due to **Telnet**'s lack of security which sends data in plain text with no encryption.

Telnet Configuration - VTY Lines

- `enable secret PASSWORD`
 - Configures a password to access privileged exec mode.
 - Required for **Telnet** to function, as you won't be able to access privileged exec mode without a password.
- `username USERNAME secret PASSWORD`
 - Configures a username and password which are used for the `login local` feature.
- `access-list ACL-ID { permit | deny } SOURCE`
 - Initializes an **ACL** which could be used to restrict connections on the Cisco device, for example.
- `line vty 0 15`
 - Enters **Virtual Teletype** configuration mode, which allows you to edit various aspects of the connection settings.
 - It is recommended to edit all VTY lines for all of them to have the same configurations.
- `login local`
 - Enables the `login local` feature, which prompts the user to enter a configured username and password before they can access the Cisco CLI.
- `exec-timeout MINUTES SECONDS`
 - Configures a timeout timer, which logs out the user after a specified period of inactivity.
- `transport input telnet`
 - Changes the protocols which are allowed to make connections to the device.
 - In this case, only **Telnet** connections are allowed.
- `access-class ACL-ID in`
 - Applies the **ACL** to the **VTY lines**.

Telnet Connection Command

To connect to a Telnet server through a PC for example, you'd need to use this command:

`telnet IP-ADDRESS`

Secure Shell (SSH) General Information

Secure Shell (SSH) is a shell connection protocol that was developed to replace less secure protocols like Telnet.

- **SSH** servers listen on port **22**.
- It provides security features such as data encryption and authentication.
- There are two **SSH** versions:
 1. **SSHv1**
 2. **SSHv2**
- If a device supports both version **1** and version **2**, it is said to run version **1.99**.

1. SSH Configuration - Check SSH Support

There are Cisco devices which support SSH and others that don't.

- IOS images that support **SSH** will have '**K9**' in their name.
- Cisco also has **NPE (No Payload Encryption)** IOS images which are sent to countries that have restrictions on encryption technologies.
 - **NPE** IOS images do not support cryptographic features such as **SSH**.

Command to Check SSH Support

You can check if a Cisco IOS image supports SSH or not through this command:

- `show version`
 - Shows information about the current device's version and OS.

2. SSH Configuration - RSA Keys

For SSH to be enabled and used, an **RSA** public and private key pair must be generated on the device. The **RSA** keys are used for **encryption/decryption**, **authentication**, and etc.

- There are two necessary conditions that must be met before **RSA** key generation is allowed on the device:
 1. The device must have a unique **hostname** (not the default hostname like '**Router**')
 2. The device must have a **DNS domain name** set up through the `ip domain name DOMAIN-NAME` command.

(Also known as a **Fully Qualified Domain Name [FQDN]**)

- When these two conditions are met, you can use the `crypto key generate rsa` command in order to generate **RSA** keys on the current device.
 - A **modulus size** must be inputted afterwards.
 - The minimum **modulus size** is **768 bits** for **SSHv2**.

3. SSH Configuration - VTY Lines

- `enable secret PASSWORD`
 - Configures a password to access privileged exec mode.
- `username USERNAME secret PASSWORD`
 - Configures a username and password which are used for the `login local` feature.
- `access-list ACL-ID { permit | deny } SOURCE`
 - Initializes an **ACL** which could be used to restrict connections on the Cisco device, for example.
- `ip ssh version 2`
 - Sets the **SSH** version to version **2**
 - Optional, but recommended
- `line vty 0 15`
 - Enters **Virtual Teletype** configuration mode, which allows you to edit various aspects of the connection settings.
 - It is recommended to edit all **VTY lines** for all of them to have the same configurations.
- `login local`
 - Enables the `login local` feature, which prompts the user to enter a configured username and password before they can access the Cisco CLI.
- `exec-timeout MINUTES SECONDS`
 - Configures a timeout timer, which logs out the user after a specified period of inactivity.
- `transport input ssh`
 - Changes the protocols which are allowed to make connections to the device.
 - In this case, only **SSH** connections are allowed.
- `access-class ACL-ID in`
 - Applies the **ACL** to the **VTY lines**.

SSH Connection Commands

To connect to an SSH server through a PC for example, you'd need to use either of these commands:

1. `ssh -l USERNAME IP-ADDRESS`
2. `ssh USERNAME@IP-ADDRESS`

Summarized Steps For SSH Configuration

For SSH to be working optimally and configured properly, you have to complete the following steps:

1. Configure a unique **hostname**
2. Configure a **DNS domain name**
3. Generate **RSA** key pairs
4. Configure an 'enable' password
5. Configure a username and password along with `login local`
6. Enable **SSHv2** (only) through the `ip ssh version 2` command
7. Configure **VTY lines**